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Systems and methods for dry eye analysis using terahertz radiation

Abstract

Methods and systems for dry eye analysis using terahertz (THz) radiation. The method includes projecting a THz wave onto a surface of an eye and detecting a reflected wave being a reflection of the THz wave reflected from the surface of the eye. The method further includes analysing properties of the eye in response to the THz wave reflected from the surface of the eye, the properties of the eye including thickness of the eye's cornea and ambient tissues, and analyzing the properties of the eye comprises measuring the thickness of the cornea and/or the ambient tissues and/or measuring an amount of chemical components of one or more of the eye's layers.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2009/0318815	12/2008	Barnes	382/128	A61B 5/444
2013/0162949	12/2012	Culjat et al.	N/A	N/A
2014/0240674	12/2013	Wei	351/208	A61B 3/102
2016/0095533	12/2015	Shang	N/A	N/A
2017/0370834	12/2016	Kassab	N/A	G01J 3/10
2018/0325431	12/2017	Guarin	N/A	A61B 5/0507
2019/0117109	12/2018	Grundfest	N/A	A61B 5/4875

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
104422702	12/2014	CN	N/A
127458	12/2017	FI	N/A
2017181201	12/2016	WO	N/A

OTHER PUBLICATIONS

Bennett et al., "Assessment of corneal hydration sensing in the terahertz band: in vivo results at 100 GHz," Journal of Biomedical Optics, Sep. 30, 2012, vol. 17, No. 9, pp. 097008-1 to 0978008-7. cited by applicant

Sung, S. et al., "Direct measurement of corneal tissue water content by reflection imaging at Terahertz Frequencies," Investigative Ophthalmology & Visual Science, Jun. 2015, vol. 56, No. 7, 1644, 2 pages. cited by applicant

Behrens, A. et al., Dysfunctional Tear Syndrome: A Delphi Approach to Treatment Recommendations, Cornea Sep. 2006, vol. 25, Issue 8, 2 pages. cited by applicant

International Preliminary Report on Patentability for PCT International Application No. PCT/SG2019/050338, issued Jan. 19, 2021, 6 pages. cited by applicant
International Search Report for PCT International Application No. PCT/SG2019/050338, mailed Sep. 2, 2019, 4 pages. cited by applicant
Lemp, M. et al., "The Definition & Classification of Dry Eye Disease," Guidelines from the 2007 International Dry Eye Workshop, Apr. 2008, 6 pages. cited by applicant
Written Opinion for Singapore Patent Application No. 11202100382Q, dated Oct. 20, 2023, 8 pages. cited by applicant
Written Opinion of the International Searching Authority for Singapore International Application No. PCT/SG2019/050338, mailed Sep. 2, 2019, 5 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This patent application is a U.S. National Phase Application Under 35 U.S.C. § 371 of International Application No. PCT/SG2019/050338, filed 10 Jul. 2019, entitled "SYSTEMS AND METHODS FOR DRY EYE ANALYSIS USING TERAHERTZ RADIATION," which claims the benefit of priority of Singapore Patent Application No. 10201806041W, filed on 13 Jul. 2018.

TECHNICAL FIELD

(2) The present invention generally relates to dry eye analysis, and more particularly relates to systems and methods for dry eye analysis using terahertz (THz) time-domain spectroscopy.

BACKGROUND OF THE DISCLOSURE

(3) The term "dry eye" includes a wide spectrum of alterations of the ocular surface with different etiology and pathophysiology. As a consequence, even the definition of dry eye is still debated, as demonstrated by the fact that in 2006 it was recommended to use the designation of "dysfunctional tear syndrome" (DTS) instead of dry eye. The International Dry Eye Workshop (DEWS) considered dry eye as an ocular surface disease, which term includes both aqueous deficient and evaporative dry eye, lid related diseases (such as meibomian gland dysfunction (MGD) and anterior blepharitis), allergic conjunctivitis and other inflammatory, infective or iatrogenic conditions.

(4) Currently available methods for diagnosis of dry eye include Schirmer's test, break-up time and ocular surface staining. Recent innovative noninvasive procedures include tear meniscus height measurement, corneal topography, functional visual acuity, tear interferometry, tear evaporimetry and tear osmolarity assessment.

(5) Dry eye analysis can be considered as cornea and ambient tissue physical property changes. Corneal dystrophies, aging and diseases, however, can complicate dry eye analysis as the physical property changes due to disease or aging may relate to the cornea as well as the ambient tissues' components' (e.g., hydration, collagen and keratocytes) changes or corneal component non-uniformity changes.

(6) Conventional methods and systems for monitoring chemical components changes in cornea as well as ambient tissues still focus on monitoring the surface of cornea as well as ambient tissues. However, to fully characterize a subject's dry eye mechanism, it is crucial and significant that the overall eye evaluation necessarily include corneal as well as ambient tissue layer thickness changes and chemical components changes resolved into individual layers.

(7) Thus, what is needed are systems and methods for dry eye analysis which evaluates corneal and

ambient tissue layer thickness changes and chemical components changes within individual layers of a subject's eye with enhanced sensitivity and selectivity. Furthermore, other desirable features and characteristics will become apparent from the subsequent detailed description and the appended claims, taken in conjunction with the accompanying drawings and this background of the disclosure.

SUMMARY

(8) According to at least one embodiment of the present invention, a system for dry eye analysis using terahertz (THz) radiation is provided. The system includes a terahertz radiation measuring system and a signal processing unit. The terahertz radiation measuring system includes a terahertz radiation generator configured to project a THz wave onto a surface of an eye and a terahertz radiation detector configured to detect a reflected THz wave reflected from the surface of the eye. The signal processing unit is configured to communicate with the terahertz radiation detector to analyse properties of the eye in response to the THz wave reflected from the surface of the eye. The properties of the eye include thickness of the eye's cornea and ambient tissues and the signal processing unit analyzing the properties of the eye includes measuring the thickness of the cornea and/or the ambient tissues and/or measuring an amount of chemical components of one or more of the eye's layers.

(9) According to another embodiment of the present invention, a method for dry eye analysis using THz radiation is provided. The method includes projecting a THz wave onto a surface of an eye and detecting a reflected wave being a reflection of the THz wave reflected from the surface of the eye. The method further includes analysing properties of the eye in response to the THz wave reflected from the surface of the eye, the properties of the eye including thickness of the eye's cornea and ambient tissues, and analyzing the properties of the eye comprises measuring the thickness of the cornea and/or the ambient tissues and/or measuring an amount of chemical components of one or more of the eye's layers.

(10) According to a further embodiment of the present invention a computer readable medium comprising instructions which, when executed by a system, make the system perform a method for dry eye analysis using THz radiation is provided. The instructions include instructions to make the system project a THz wave onto a surface of an eye and detect a reflected THz wave reflected from the surface of the eye. The instructions further include instructions to make the system analyse properties of the eye in response to the THz wave reflected from the surface of the eye, the properties of the eye comprising thickness of the eye's cornea and ambient tissues, wherein analyzing the properties of the eye includes measuring the thickness of the cornea and/or the ambient tissues and/or measuring an amount of chemical components of one or more of the eye's layers.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views and which together with the detailed description below are incorporated in and form part of the specification, serve to illustrate various embodiments and to explain various principles and advantages in accordance with a present embodiment.

(2) FIG. 1 depicts an illustration of a terahertz radiation measuring system in accordance with present embodiments.

(3) FIG. 2 depicts a flowchart of a method for dry eye analysis in accordance with the present embodiments using the terahertz radiation measuring system of FIG. 1.

(4) FIG. 3 depicts an illustration of a system for dry eye analysis in accordance with the present

embodiments using the terahertz radiation measuring system of FIG. 1.

(5) FIG. 4 depicts a planar cross-sectional view of a smart box interface for use with the system of FIG. 3 to enhance sensitivity and/or selectivity of dry eye analysis of an eye in accordance with the present embodiments.

(6) FIG. 5 depicts a graph of terahertz radiation signals of hydroxide bonds and free liquid water at different concentrations as measured by the system of FIG. 3 in accordance with the present embodiments.

(7) FIG. 6 depicts a graph of monitoring of cultured collagen samples by terahertz radiation signals using the system of FIG. 3 in accordance with the present embodiments.

(8) FIG. 7 depicts an illustration of layers and layer interfaces of the cornea of an eye.

(9) FIG. 8, comprising FIGS. 8A, 8B, 8C and 8D, depicts an illustration of how sublayer properties can be extracted non-invasively by focusing terahertz radiation through the epithelial layer and into the stroma layer of the eye, wherein FIG. 8A an illustration of focusing the terahertz radiation through the epithelial layer and into the stroma layer of the eye, FIG. 8B depicts an illustration of stripping of the epithelial layer and focusing the terahertz radiation through the stroma surface and into the stroma layer of the eye, FIG. 8C depicts a graph of results of the terahertz radiation focusing of FIG. 8A, and FIG. 8D depicts a graph of results of the terahertz radiation focusing of FIG. 8B.

(10) FIG. 9, comprising FIGS. 9A and 9B, depicts graphs of properties of an epithelial layer of the eye as terahertz radiation is focused through the epithelial layer in accordance with the present embodiments using the system of FIG. 3, wherein FIG. 9A depicts a graph of time domain spectra as the terahertz radiation beam is focused from the epithelium surface down through the epithelium to the epithelium stroma interface, and FIG. 9B depicts a graph of frequency domain spectra as the terahertz radiation beam is focused from the epithelium surface down through the epithelium to the epithelium stroma interface.

(11) FIG. 10, comprising FIGS. 10A and 10B, depicts graphs of frequency domain characteristics of the epithelial layer of the eye in accordance with the present embodiments as derived from the frequency domain spectra of FIG. 9B, wherein FIG. 10A depicts a graph of the frequency domain spectra of FIG. 9B when fast fourier transform (FFT) is performed on data only for the first peak in the time domain spectra of FIG. 9A, and FIG. 10B depicts a graph of the frequency domain spectra of FIG. 9B when fast fourier transform (FFT) is performed on data only for the second peak in the time domain spectra of FIG. 9A.

(12) FIG. 11 depicts a graph of time domain spectra for corneal layer thickness measurement by the system of FIG. 3 in accordance with the present embodiments.

(13) And FIG. 12, comprising FIGS. 12A and 12B, depicts graphs comparing time domain spectra results using terahertz radiation in accordance with the present embodiments of a set of five porcine eyes with ocular corneal thickness (OCT) measurements using conventional systems of the set of five porcine eyes, wherein FIG. 12A is a graph of the time domain spectra results using terahertz radiation in accordance with the present embodiments and FIG. 12 B is a graph of the OCT measurements using conventional systems.

(14) Skilled artisans will appreciate that elements in the figures are illustrated for simplicity and clarity and have not necessarily been depicted to scale.

DETAILED DESCRIPTION

(15) The following detailed description is merely exemplary in nature and is not intended to limit the invention or the application and uses of the invention. Furthermore, there is no intention to be bound by any theory presented in the preceding background of the invention or the following detailed description. It is the intent of the present embodiment to present rapid, in vivo, non-invasive, non-destructive and non-contact systems and methods for dry eye tissue analysis which evaluate corneal and ambient tissue layer thickness changes and chemical components changes within individual layers of a subject's eye with enhanced sensitivity and selectivity.

(16) According to the definition of dry eye adopted by the International Dry Eye Workshop (DEWS) in 2007, dry eye is “a multifactorial disease of the tears and ocular surface that results in symptoms of discomfort, visual disturbance and tear film instability with potential damage of the ocular surface. It is accompanied by increased osmolarity of the tear film and inflammation of the ocular surface.”

(17) Monitoring chemical component changes in the cornea as well as in ambient tissues is a method for analysis of the root reasons of dry eye formation. Therefore, dry eye analysis includes analysis of cornea and overall tissue physical property changes in layers of the eye. Corneal dystrophies, aging and diseases, however, can complicate dry eye analysis. The physical property changes due to disease or aging could very easily relate to corneal as well as ambient tissues' component changes such as hydration, collagen and keratocyte changes or to corneal component non-uniformity changes.

(18) In accordance with present systems and methods, an accurate in-vivo instant terahertz measurement system is provided to extract in-vivo cornea as well as ambient tissues' component properties and chemical composition (such as hydration, collagen and keratocytes) and give information of the chemical component distribution vertically in order to analyse and monitor dry eye formation. While some conventional methods to evaluate dry eyes monitor changes of tear secretion or tear film then evaluate dry eye formation by evaluation of the tear film offline, the method in accordance with the present embodiments focuses on in vivo non-invasive and non-destructive analysis of the dry eye tissues.

(19) Typical corneal thickness measurement is accomplished by optical coherence tomography, ultrasound, or interference of two light rays. For cornea as well as other eye ambient tissues' chemical component extraction (e.g., analysis of collagen, hydration and keratocytes composition), there are no reliable acknowledged methods.

(20) Present embodiments address the drawbacks of prior art systems and methods by providing an in vivo non-invasive technology for measuring properties of the eye such as thickness of the eye's cornea and ambient tissues and/or measuring an amount of chemical components of one or more of the eye's layers. Embodiments of systems and methods of the present invention operate in the THz range in the electromagnetic spectrum which lies between microwave and infrared frequencies and generally defines frequencies ranging from 100 GHz (10^{sup}.11 Hz, 3 mm wavelength) to 10 THz (10^{sup}.13 Hz, 3.3 μm wavelength). Electromagnetic radiation in the THz range may also be referred to as THz light, THz radiation, or THz waveforms.

(21) In accordance with present embodiments, a terahertz radiation measuring system, such as a terahertz portable system, analyses dry eye conditions in-vivo using terahertz time-domain spectroscopy. Referring to FIG. 1, an illustration **100** of a system for analysing dry eye conditions of an eye using terahertz (THz) radiation in accordance with present embodiments uses a THz time-domain (THz-TDS) system **102** to provide an illumination beam of terahertz radiation and a computer or other processing means **104**.

(22) The THz-TDS system **102** can be configured to reflect electromagnetic radiation **110** in THz range from a THz transmitter or emitter **112** toward a surface of an eye **114**, receive THz light reflected from the eye at a THz detector or receiver **116**, and generate a signal **118** indicative of the received radiation which is amplified and digitized by circuitry **120**. The THz radiation is generated by the THz transmitter **112**, such as an antenna or a nonlinear crystal, and pulsed in response to a signal from a femtosecond laser **122**. The THz detector or receiver **116** can also be an antenna or a nonlinear crystal. The computer **104**, which communicates with the THz-TDS system **102**, can be configured to process the generated signal **118** and may further be configured for creating a visual imaging of the THz response from the surface or sublayers of the eye **114**. A XYZ stage **105** can be configured to manipulate the structure of the THz-TDS system **102** to scan the surface of the eye **114** and move THz focal point **124** from a surface of the eye **114** to a level below the surface of the eye.

(23) Thus, the system **100** for dry eye analysis includes an illumination system (e.g., the femtosecond laser **122** and the THz transmitter **112**) configured to provide an illumination beam of terahertz radiation **110**, an optical system including mirrors and lenses (e.g., lens **113**) arranged in an optical path of the terahertz radiation **110** to relay and direct at least a portion of the illumination beam of terahertz radiation **110** onto an eye **114** of a subject and to receive at least a portion of terahertz radiation reflected from the eye to provide a return beam of terahertz radiation, and a detection system (e.g., detector **116**) arranged in an optical path of said return beam of terahertz radiation, the detection system configured to provide a detection signal **118** from detecting at least a portion of said return beam of terahertz radiation. A signal processing system **120** is configured to communicate with the detection system **116** to receive the detection signal **118**, wherein the signal processing system processes the detection signal to provide a measure of elastic and rigidity in the cornea **114** of the subject.

(24) As seen from FIG. **1**, the illumination system comprises a THz generator head including the femtosecond laser **122** and a nonlinear optical crystal (i.e., the THz transmitter **112**). The optical system is arranged in an optical path of the THz illumination system to relay and focus oblique-angle illumination of terahertz radiation on the subject's eye **114** at the focal point **124**. The illumination optical system, which can be open space or fiber connected, includes a pair of off-axis parabolic mirrors **126**, **128** and a pick-up optical system configured to provide the return beam of terahertz radiation also includes a pair of off-axis parabolic mirrors **130**, **132**. The detection system **116** is configured to detect said return beam of terahertz radiation within a frequency band of about 0.1 THz to about 10 THz.

(25) The system **100** for dry eye analysis measures the cornea as well as the ambient tissues thickness and the chemical component extraction at individual layers of the eye. The THz radiation measuring system **100** incorporates a small robust femtosecond laser head (the femtosecond laser **122**) and a fast optical delay line **134** enabling a sampling signal at or larger than 150 Hz per second (i.e., 6-7 milliseconds). Using systems and methods in accordance with the present embodiments, patients can be scanned without even noticing the scanning.

(26) FIG. **2** depicts a flowchart **300** of an exemplary method for dry eye analysis in accordance with the present embodiments. The method includes projecting THz radiation onto the eye's cornea as well as ambient tissues of the eye **202**. The reflected THz signal reflected from the cornea and/or the ambient tissues of the eye and arriving at the detector **116** is measured **204**. Then, the beam of terahertz radiation **110** is defocused and refocused to a subsurface/sublayer of the cornea and/or ambient tissue and measured a second time **206**. The signal processing unit **120** processes the terahertz radiation signal **118** to determine **208** properties of the cornea and/or the ambient tissues of the eye. The thickness of layers and the properties (e.g., amount of chemical components) of the layers are determined **210** by the computer **104**. Finally, the dry eye conditions are evaluated **212** based on the chemical composition distribution in the cornea and in the ambient tissues of the eye.

(27) One of objectives of the present embodiments is to provide a non-contact, in vivo, non-invasive system for analyzing and evaluating dry eye conditions of an eye. Referring to FIG. **3**, an illustration **300** depicts use of the system **100** for performing dry eye analysis of a subject's eye **114** in accordance with the present embodiments. To achieve the above objective, the present embodiments provide a non-invasive system for measuring properties of the subject's eye **114** including measuring the thickness of the cornea and/or the ambient tissues and/or measuring an amount of chemical components of one or more of the eye's layers. The terahertz radiation measuring system (THz-TDS) includes the THz wave generator **112** and the THz detector **116** and generates terahertz radiation before and after defocusing and refocusing to measure cornea and ambient tissue properties. The signal processing unit **120** includes a computer **202** which controls the terahertz radiation measuring system (THz-TDS), as well as analysing properties of the eye in response to the THz wave reflected from the surface of the eye. The computer **202** includes a user friendly interface to control the THz source and detection, to analyse the data and do the necessary

corrections and display the results and graphs

(28) In accordance with the present embodiments, the chemical components of the eye's layers for analyzing dry eye include chemical components such as collagen, hydration and keratocytes. The computer **202** analyses the properties of the eye by determining the properties of one or more layers of the eye, such as calculating a relative corneal composition of the chemical components in the eye's cornea or determine an absolute corneal composition of the chemical components at positions of different cornea layer interfaces using a predefined calibration method, where the different cornea layer interfaces include an interface between the eye's cornea and a layer selected from an epithelium layer, a stromal layer, and an endothelium layer. The absolute corneal composition of the chemical components may include a point value that is an absolute corneal hydration for at least one local region of the eye's cornea. Furthermore, the computer **202** may calculate an overall average of one or both of the thickness or the chemical components of the eye's cornea in response to the THz wave reflected from the surface of the eye using a predetermined model. Thus, the system **300** is able to differentiate the cornea layers (epithelium, epithelium/stromal interface, stromal as well as stromal/endothelium interface) and extract the components (hydration, collagen, keratocytes) in terms of layer.

(29) The THz band represents a unique combination of relatively high spatial resolution with the low scatter and large interaction with chemical bonding at the molecular level. Thus, THz radiation has become a powerful tool for assessing chemical molecular materials. THz spectroscopy can identify an amount absorption fingerprint of the cornea and ambient tissue chemical components (e.g., collagen and keratocytes) by monitoring fingerprint peak positions, intensities and phase changes, which can be correlated to physical structure changes related to dry eye.

(30) The present embodiments use a THz pulse to detect eye ambient tissues chemical components, different cornea layers and interfaces. Through fast fourier transform (FFT) analysis of frequency domain spectra, absorption peaks of water, collagen and keratocytes can be identified. Chemical components of the cornea at individual layers as well as of ambient tissues can be extracted. While a conventional cornea hydration sensing has been reported, it is unrelated to dry eye analysis and such sensing could only detect surface hydration level without other chemical components analysis and did not disclose any analysis of individual interfaces or different layers of the cornea and also did not disclose any analysis of layer thickness.

(31) The present embodiments are technologically significant in the way that they allow extraction of cornea as well as ambient tissue chemical component information in accordance with different layers instantly (i.e., in milliseconds), non-invasively, and non-destructively without the patient sensing the measurements being performed.

(32) Referring to FIG. 4, a planar cross-sectional view **400** depicts a smart box interface **402** for use with the system **300** to enhance sensitivity and/or selectivity of dry eye analysis of an eye **114** in accordance with present embodiments. The XYZ stage **105** is configured to hold the THz structure, scan the eye **114** and move the THz focus point **124** to different places/layers of the eye **114**. To enhance the sensitivity and selectivity of the collagen and keratocyte detection signals in the THz range, a smart box **404** can be inserted between the beam of THz radiation **110** and the eye **114**. The smart box **404** can also offset hydration effects from the environment by using hydration absorption materials or low refractive index materials such as polyethylene.

(33) The computer **202** communicates with THz-TDS measuring system **100** and can be configured to process the generated signal **118**. The computer **202** may be further configured to create a visual image of the THz response. The XYZ stage **105** can be coupled to the computer **202** to move THz focal point **124** to scan the eye **114** and scan different layers of the eye **114**.

(34) Due to the presence of water in physiological tissue and the high THz absorption of water, reflective THz imaging has distinct advantages over transmission-based systems, especially for in vivo applications. The dielectric properties of water absorption frequencies yield easily detectable changes in THz reflectivity. Small changes in hydration levels demonstrate an effective contrast

mechanism in THz spectral. These advantages coupled with the low, non-ionizing THz photon energy (i.e., 0.4-40 meV) makes THz radiation an ideal tool for in vivo imaging of skin burns, melanoma/carcinoma, corneal pathologies, and cancers as well as for in vivo dry eye analysis. (35) Referring to FIG. 5, a graph **500** depicts terahertz radiation signals of hydroxide bonds and free liquid water at different concentrations as measured by the system **300** in accordance with the present embodiments. Hydration-sensitive data and imagery in many test targets is a mixture of chemical and liquid water. Table 1 identifies the chemicals propanol, ethanol and methanol and the graph **500** depicts deionized water mixed with these chemicals at different concentrations.

(36) TABLE-US-00001 TABLE 1 Compound IUPAC Name Common Name CH.sub.3OH Methanol Methyl alcohol CH.sub.3CH.sub.2OH Ethanol Ethyl alcohol CH.sub.3CH.sub.2CH.sub.2OH 1-Propanol Propyl alcohol

(37) Both THz peak intensities and peak positions show good correlation with the water concentration. The in-set graph **502** shows the frequency domain spectral where the peaks indicate water absorption. As water mixing with collagen and keratocytes with different concentrations will also show trendy spectral, water concentrations in the cornea and the ambient tissues of the eye can be calculated. By comparing the measured samples with standard samples, a method is prescribed to extract absolute hydration, collagen and keratocytes concentration levels. In addition, ex vivo corneas such as porcine corneas, in a variety of configurations and hydration states can be used to define the relationship between corneal hydration and terahertz reflectivity.

(38) Referring to FIG. 6, a graph **600** depicts monitoring of cultured collagen samples by terahertz radiation signals using the system **300** in accordance with the present embodiments. A series of pure cultured collagen sheets were tested using the THz-TDS system **300**. The intensity **602** vs. frequency **604** is graphed for the cultured collagen samples at five minutes **610**, ten minutes **620** and fifteen minutes **630**. The obvious observation is that the peak **640** at 0.065 THz is stable. This collagen peak **640** is discernible in THz radiation measurement of chemical components in the cornea and/or ambient tissues of the eye.

(39) Referring to FIG. 7, an illustration **700** depicts layers and layer interfaces of the cornea of an eye. Epithelial tissues line the outer surfaces of organs and blood vessels. The epithelium **710** is the outer surface of the cornea of the eye. An interface, the Bowman's membrane **720**, separates the epithelium **710** from the stroma **730**. The Descement's membrane **740** is an interface between the stroma **730** and the endothelium **750**.

(40) In order to determine the interference between surfaces, layers and interfaces when measuring properties of the layers with THz radiation, the epithelium and stroma layers of porcine eyes were investigated. Referring to FIGS. 8A and 8B, illustrations **800**, **820** depict focusing the terahertz radiation **110** through the epithelial layer **802** and into the stroma layer **804** of the eye (the illustration **800**) and focusing the terahertz radiation **110** through the stroma surface **810** and into the stroma layer **804** of the eye where the epithelial layer **802** has been stripped off the stroma layer **804** (the illustration **820**). Referring to FIGS. 8C and 8D, the graphs **840**, **860** illustrate how sublayer properties can be extracted non-invasively by focusing terahertz radiation through the epithelial layer and into the stroma layer of the eye. The graph **840** depicts the THz-TDS spectra collected from reflection of the terahertz radiation **110** focusing of the illustration **800** and the graph **860** depicts the THz-TDS spectra collected from reflection of the terahertz radiation **110** focusing of the illustration **820**. It can be seen that with and without the epithelium layer **802**, the 0.2 to 0.4 THz peak **842**, **862** in the deep in focus **844**, **864** and the 0.2 to 0.4 THz peak **846**, **866** in the stromal surface focus **848**, **868** in both graphs **840**, **860** shows the same peak position and intensity for the stromal information. This shows that by focusing the THz radiation **110** through surface with the epithelium **802** intact, the stromal layer **804** chemical component information can be directly detected. The 0.067 THz peaks **850**, **870** are from collagen components in the stromal layer **804**. The peak positions **850**, **870** remain unchanged with or without the epithelial layer **802**. The collagen peaks **850**, **870** can thus be used as to analyze the corneal health as well as used as an

indicator to extract other information such as indirectly extract corneal hydration level.

(41) In order to successfully observe the fingerprint of the cornea, the samples were covered using a simple paperbox and, while the THz curves show decreased air-water vapour effects, the time domain and frequency domain signals show much smoother curves. FIGS. **9A** and **9B** depict graphs **900**, **950** of properties of an epithelial layer of the eye as terahertz radiation is focused through the epithelial layer in accordance with the present embodiments using the system **300**. The graph **900** depicts a time domain spectra **910** of the THz radiation focused on the epithelium surface and a time domain spectra **915** of the THz radiation focused on the epithelium-stroma interface. The graph **950** depicts a frequency domain spectra **960** of the THz radiation focused on the epithelium surface and a frequency domain spectra **965** of the THz radiation focused on the epithelium-stroma interface. As the terahertz radiation beam is focused from the epithelium surface down through the epithelium to the epithelium stroma interface, the time domain spectra shifted from **910** to **915**. The second peaks shifted from **912** to **917** which indicate a second interface. The epithelium/stromal interface and the stromal quality is crucial information input for successful operation rate and minimized post-surgery complications for LASIK correction surgery as the anterior stromal is the site cut during LASIK.

(42) From the graph **950**, in the frequency domain when focused on the surface and epithelium/stromal interface, some character peaks **962**, **967** can clearly be identified. However, these peaks are likely water vapour absorption peaks and collagen/keratocytes fingerprint peaks.

(43) In order to identify each layers' information, two main peaks **914**, **919** which represent the epithelium surface and epithelium/stromal interface are identified in the time domain graph **900**. Fast Fourier transform (FFT) is performed separately in order to observe the frequency domain characteristic peaks separately. FIGS. **10A** and **10B** depict graphs **1000**, **1050** of frequency domain characteristics of the epithelial layer of the eye in accordance with the present embodiments as derived from the frequency domain spectra of the graph **950**. The graph **1000** depicts the frequency domain spectra **960** when FFT is performed on data only for the first peak **914** (FIG. **9A**) and the graph **1050** depicts the frequency domain spectra **965** when FFT is performed on data only for the second peak **919** (FIG. **9A**). Obvious fingerprint peaks **1052**, **1054** can be identified at 86 GHz, peaks **1058**, **1060** at 1.1 THz, peaks **1064**, **1066** at 1.4 THz, and peaks **1070**, **1072** at 1.65 THz can be easily identified. From these results, the epithelium layer hydration level and components can be separately analysed. The stromal layer hydration level, collagen and keratocytes distribution can also be analysed directly.

(44) Another objective of the present embodiments is to provide a rapid and non-contact method to measure corneal thickness. To achieve this objective, the systems and methods in accordance with the present embodiments utilize the reflected terahertz signals from a front surface of a cornea and a rear interface of the cornea to determine the corneal thickness. The amount of reflection from the rear surface of the cornea is small compared with the amount of reflection from the front surface of the cornea. When reflected terahertz radiation is received by the detector **116** (FIG. **3**), the peak of the reflected THz from the rear surface is smaller than that from the front surface of the cornea. The cornea thickness is then related to the time delay of the two signals and the refractive index of the cornea. The greater the time delay is, the thicker the cornea layer is. Furthermore, the each individual layers of the corneal can be extracted from the time of flight between peaks of sub interfaces. FIG. **11** depicts a graph **1100** of time domain spectra for corneal layer thickness measurement by the system of **300** in accordance with the present embodiments. The time of flight between the 2.sup.nd and 3.sup.rd peaks is correlated to the thickness of the stromal layer. The precise thickness can be corrected further based on the optical parameters derived.

(45) FIGS. **12A** and **12B** depict graphs **1200**, **1250** comparing time domain spectra results using terahertz radiation in accordance with the present embodiments of a set of five porcine eyes with ocular corneal thickness (OCT) measurements using conventional systems of the set of five porcine eyes. The graph **1200** depicts a time domain spectra results using terahertz radiation in accordance

with the present embodiments and the graph **1200** depicts the OCT measurements using conventional systems. It can be seen that the THz spectra results has almost the same trend as the OCT results. The OCT results are acknowledged to be a proximal reflection of the corneal hydration level. However, the OCT suffers serious deviation from the real hydration level of the cornea. The THz spectra advantageously provides a more precise and more reliable value of corneal hydration level as compared with the OCT results. Yet, the similar trends given out by THz TDS in the graph **1200** and the OCT in the graph **1250** prove that THz radiation in the systems and methods in accordance with the present embodiments can identify the properties of the eye.

(46) The present embodiments are technologically significant in the way that they allow extraction of cornea as well as ambient tissue chemical component information in accordance with different layers instantly (i.e., in milliseconds), non-invasively, and non-destructively without the patient sensing the measurements being performed. Thus, it can be seen that the present embodiments provide rapid in vivo non-invasive systems and methods for dry eye analysis. The non-destructive and non-contact systems and methods in accordance with the present embodiments evaluate corneal and ambient tissue layer thickness changes and chemical components changes within individual layers of a subject's eye with enhanced sensitivity and selectivity for dry eye tissue analysis.

(47) While exemplary embodiments have been presented in the foregoing detailed description of the invention, it should be appreciated that a vast number of variations exist. It should further be appreciated that the exemplary embodiments are only examples, and are not intended to limit the scope, applicability, operation, or configuration of the invention in any way. Rather, the foregoing detailed description will provide those skilled in the art with a convenient road map for implementing an exemplary embodiment of the invention, it being understood that various changes may be made in the function and arrangement of steps and method of operation described in the exemplary embodiment without departing from the scope of the invention as set forth in the appended claims.

Claims

1. A system for dry eye analysis using terahertz (THz) radiation, the system comprising: a terahertz radiation measuring system comprising: a terahertz radiation generator configured to project a THz wave onto a surface of an eye, the terahertz radiation generator comprising an optical delay line configured to provide a sampling rate of the THz wave of at least 150 Hz; a stage configured to focus the THz wave to the surface of the eye or a layer below the surface of the eye; and a terahertz radiation detector configured to detect a reflected wave being a reflection of the THz wave reflected from the surface of the eye or a layer below the surface of the eye; a smart box interface between the THz wave and the eye, the smart box interface comprising a smart box comprising low refractive index material and/or hydration absorption material for enhancement of sensitivity or selectivity of analysis of properties of the one or more layers of the eye; and a signal processing unit comprising a computing device, the signal processing unit configured to communicate with the terahertz radiation detector to analyse the properties of one or more layers of the eye in response to the THz wave reflected from the surface of the eye or a layer below the surface of the eye, the properties of the one or more layers of the eye comprising a thickness of the eye's cornea and ambient tissues and an amount of a plurality of chemical components in the one or more layers of the eye, wherein the signal processing unit is further configured to extract the analyzed properties of the one or more layers of the eye; wherein the computing device of the signal processing unit is configured to process a signal representing the reflected wave to determine the thickness of the cornea and the thickness of the ambient tissues of the one or more layers of the eye and the amount of the plurality of chemical components of the one or more layers of the eye to generate an evaluation of the dry eye conditions of the eye, and to present results of the evaluation of the dry

eye conditions of the eye and visual imaging of the analyzed properties of the cornea or the one or more layers of the eye.

2. The system in accordance with claim 1 wherein the chemical components of the eye's layers comprise a chemical component selected from collagen, hydration and keratocytes.

3. The system in accordance with claim 1 wherein the computing device is configured to calculate a relative corneal composition of the chemical components in the eye's cornea.

4. The system in accordance with claim 1 wherein computing device is configured to determine an absolute corneal composition of the chemical components at positions of different cornea layer interfaces using a predefined calibration method, and wherein the different cornea layer interfaces comprise an interface between the eye's cornea and a layer selected from an epithelium layer, a stromal layer, and an endothelium layer.

5. The system in accordance with claim 4 wherein the computing device is further configured to calculate a point value that is an absolute corneal hydration for at least one local region of the eye's cornea.

6. The system in accordance with claim 1 wherein the computing device is configured to calculate an overall average of one or both of the thickness or the chemical components of the eye's cornea in response to the THz wave reflected from the surface of the eye or a layer below the surface of the eye using a predetermined model.

7. The system in accordance with claim 1 wherein the terahertz radiation measuring system further comprises an illumination optical system coupled to the terahertz radiation generator to provide oblique-angle terahertz radiation illumination of the surface of the eye and coupled to the terahertz radiation detector to provide the reflected wave thereto, and wherein the illumination optical system comprises optical components connected by open space or optical fiber.

8. The system in accordance claim 7 wherein the illumination optical system comprises two off-axis parabolic (OAP) mirrors, wherein a first mirror of the two OAP mirrors is arranged in an optical path of the terahertz radiation generator and is configured to receive the THz wave emitted from the terahertz radiation generator and reflect at least a portion of the THz wave and project the at least a portion of the THz wave onto the surface of the eye; and wherein a second mirror of the two OAP mirrors is arranged in a symmetrical position to the first mirror with respect to the eye and is configured to receive at least a portion of the THz wave reflected from the eye.

9. The system in accordance with claim 1 wherein the terahertz radiation detector is configured to detect the reflected wave within a frequency band of 0.1 THz to about 10 THz.

10. The system in accordance with claim 1 wherein the terahertz radiation generator further comprises: a femtosecond laser configured to generate the THz wave; and a terahertz radiation signal emitter configured to emit the THz wave onto the surface of the eye, wherein the terahertz radiation signal emitter is an antenna or a nonlinear crystal.

11. The system in accordance with claim 1 wherein the computing device comprises a user interface configured to display the visual imaging of the analyzed properties of the cornea or the one or more layers of the eye.

12. The system in accordance with claim 11 wherein the computing device is further coupled to the terahertz radiation measuring system, and wherein the user interface is further configured to enable control of the terahertz radiation measuring system.

13. A method for dry eye analysis using terahertz (THz) radiation, the method comprising: projecting a THz wave onto a surface of an eye such that a sampling rate of the THz wave is at least 150 Hz; focusing the THz wave to the surface of the eye or a layer below the surface of the eye; detecting a reflected wave being a reflection of the THz wave reflected from the surface of the eye or a layer below the surface of the eye as a result of projecting the THz wave onto the surface of the eye; and determining properties of one or more layers of the eye in response to the THz wave reflected from the surface of the eye or a layer below the surface of the eye, the properties of the one or more layers of the eye comprising a thickness of the eye's cornea and ambient tissues and an

amount of a plurality of chemical components in the one or more layers of the eye; extracting the properties of the one or more layers of the eye; analyzing the extracted properties of the one or more layers of the eye to generate an evaluation of the dry eye conditions of the eye; and presenting the evaluation of the dry eye conditions of the eye and visual imaging of the analyzed properties of the cornea or the one or more layers of the eye, wherein projecting the THz wave onto the surface of the eye comprises disposing a smart box interface between the THz wave and the eye, the smart box interface comprising a smart box comprising low refractive index material and/or hydration absorption material for enhancement of sensitivity or selectivity of analysis of the properties of the one or more layers of the eye.

14. The method in accordance with claim 13 wherein the chemical components of the eye's layers comprise a chemical component selected from collagen, hydration and keratocytes.

15. The method in accordance with claim 13 wherein determining the properties of the one or more layers of the eye comprises calculating a relative corneal composition of the chemical components in the cornea of the eye.

16. The method in accordance with claim 13 wherein determining the properties of the one or more layers of the eye comprises determining an absolute corneal composition of the chemical components at positions of different cornea layer interfaces using a predefined calibration method, wherein the different cornea layer interfaces comprise an interface between the eye's cornea and a layer selected from an epithelium layer, a stromal layer, and an endothelium layer.

17. The method in accordance with claim 16 wherein determining the absolute corneal composition of the chemical components comprises calculating a point value that is an absolute corneal hydration for at least one local region of the eye's cornea.

18. The method in accordance with claim 17 wherein determining the absolute corneal composition of the chemical components comprises calculating an overall average of one or both of the thickness or the chemical components of the eye's cornea in response to the THz wave reflected from the surface of the eye or a layer below the surface of the eye using a predetermined model.
